Ella Lorraine Wei

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EDUCATION

Texas A&M University — *College Station, TX* May 2029
B.S. in Mechanical Engineering | Minors: Math & AI in Business GPA: 3.689 | Engineering Honors
Relevant Coursework: Geometric Modeling for Mechanical Design, Engineering Mechanics, Differential Equations

WORK EXPERIENCE

Worcester Polytechnic Institute **May 2026 – August 2026**
Transportation Research Assistant Worcester, MA (Remote)

- Evaluated Pittsburgh transit survey demographics against Census benchmarks, identifying an 18% discrepancy to assess sampling bias and data reliability.
- Leveraged Claude for AI-assisted workflow development, troubleshooting Excel formulas and refining analytical approaches while independently verifying outputs and research.

Autonomous Systems Laboratory **May 2026 – Present**
Undergraduate Research Assistant College Station, TX

- Utilize the Robot Operating System (ROS) to develop, test, and evaluate autonomous robotic systems and software.
- Develop and evaluate robotic systems and experimental methodologies while collaborating with undergraduate and graduate researchers to analyze results, document findings, and communicate technical progress.

Texas A&M University **August 2026 – Present**
Peer Teacher | ENGR102 & PHYS216/ENGR216 College Station, TX

- Collaborate with course instructors and teaching staff to support students in ENGR 102 and PHYS 216/ENGR 216.
- Explain complex engineering concepts through clear, accessible instruction while guiding students through technical problem-solving and applications, adapting explanations to individual learning styles and levels of understanding.

PROJECT EXPERIENCE

PRNT: Non-Planar 3D Printer **September 2026 – Present**
TURTLE Robotics | Member College Station, TX

- Contribute to the mechanical design and prototyping for a 4-axis R-θ-Z-A 3D printer capable of non-planar extrusion.
- Develop and evaluate CAD models and mechanical mechanisms for additional rotational axes, considering kinematics, manufacturability, and integration with existing 3D-printing hardware.
- Collaborate with a multidisciplinary team on non-planar printing methods aimed at reducing support material and optimizing anisotropic material properties

Swim Shady: Autonomous Underwater Vehicle **September 2026 – Present**
WIRED Robotics | Mechanical, Electrical, & Business Subteam Member College Station, TX

- Contribute to the design and development of an autonomous underwater vehicle (AUV) for the annual RoboSub competition across mechanical, electrical, and business functions.
- Support mechanical design, CAD, electrical integration (PCP, soldering, etc.), and system testing while collaborating across subteams to develop and troubleshoot the vehicle.

ACTIVITIES & LEADERSHIP

Institute for the Development and Education for Asian American Leaders **August 2025 – Present**
Mentor | Public Relations Chair College Station, TX

- Mentor first-year APIDA students & lead public relations efforts through newsletters, graphics, and social media, growing organizational reach by 500+ followers while supporting mentorship and professional development events.

Houston SynchroStars **May 2017 – May 2026**
Team Leader | Novice Team Coach Sugar Land, TX

- Competed as an international artistic swimmer, contributing to a 2023 Junior Olympics overall championship and earning 1st in Individual Figures and 2nd in Youth Duet at the 2022 Youth World Championships.

HONORS & AWARDS

VEX Robotics World Championship Qualifier | VEX Robotics State Champion | 2023 Junior Olympics Team Overall Champion | 2022 Youth World Championships Individual Figure 1st, Youth Duet 2nd | USACO Silver Division | Unity Certified

TECHNICAL SKILLS

Languages | Robot Operating System, Java, Python, C#, Unity, HTML/CSS/JS, LaTeX
Software | SolidWorks, AutoCAD, Microsoft Excel, Microsoft Word, Microsoft Powerpoint, Adobe After Effects, Google
Developer Tools | Git, Visual Studio Code, PyCharm, Eclipse, Netbeans, Spyder, Github, IntelliJ